PRECALCULUS · UNIT 3: CONICS

Science Museum Conics

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design a science museum wing. Circles describe the planetarium, parabolas focus the satellite dish and flashlight, ellipses create the whispering gallery, and hyperbolas model a cooling tower and navigation.

Locating foci gives every equation a physical purpose.

Standards Addressed

- G-GPE.A.1 Circle equations; completing the square
- G-GPE.A.2 Parabola from focus and directrix
- G-GPE.A.3 Ellipses and hyperbolas from foci (+)
- Systems involving conics

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with photos of a whispering gallery. Part 1 circles and parabolas.
Day 2	Part 2 ellipses and foci.
Day 3	Part 3 hyperbolas and the exhibit packet.

Materials

- Student handout and Specification Cards
- Graphing calculators or Desmos
- String, pins, and cardboard to draw ellipses

How the Three Versions Differ

Level	What changes
CP2	Circle equations, focus of $x^2 = 4py$, finding p from a point, and identifying conics.
CP1	Completing the square, focal lengths, ellipse foci and eccentricity, and hyperbola navigation.
Honors	Deriving the parabola, arch clearance, ellipse redesign, conic intersections, and a navigation fix.

Watch For (Common Misconceptions)

- Sign errors in $(x - h)^2$ for the center.
- Using $c^2 = a^2 + b^2$ for an ellipse.
- Confusing a (vertex distance) with c (focus distance).

Extension Ideas

- Draw an ellipse with two pins and string.
- Research real whispering galleries.
- Build a parabolic solar cooker model.

Answer Key – CP2

Part 1: Planetarium Circle

#	Task	Answer
1	Write the equation of the planetarium floor (center (0, 0), radius 25 ft).	$x^2 + y^2 = 625$
2	Is the point (15, 20) on the edge of the floor? Show your check.	225 + 400 = 625, yes
3	The projector is moved to (3, −2). Write the equation of a circle of radius 25 centered there.	$(x - 3)^2 + (y + 2)^2 = 625$
4	Find the center and radius of $(x + 1)^2 + (y - 4)^2 = 49$.	Center (−1, 4), radius 7

Part 2: Parabolic Dish

#	Task	Answer
5	The flashlight mirror is $y = \frac{1}{8}x^2$, or $x^2 = 8y$. Find p and the focus.	$4p = 8 \rightarrow p = 2$; focus (0, 2)
6	Where should the bulb go? Why?	At the focus (0, 2); light from the focus reflects into a parallel beam.
7	The dish passes through (6, 1.5). Find p using $x^2 = 4py$.	$36 = 6p \rightarrow p = 6$; the receiver is 6 ft above the vertex
8	Write the equation of the dish.	$x^2 = 24y$

Part 3: Ellipse & Hyperbola

#	Task	Answer
9	Identify the conic: $\frac{x^2}{900} + \frac{y^2}{400} = 1$. What are its length and width?	Ellipse; 60 ft long and 40 ft wide
10	Identify the conic: $\frac{x^2}{225} - \frac{y^2}{400} = 1$.	Hyperbola (minus sign)
11	Sketch your favorite exhibit and label its key points.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why is the receiver of a satellite dish placed at the focus?	Signals that arrive parallel to the axis reflect to the focus.
R2	How can you tell a hyperbola from an ellipse in an equation?	An ellipse adds the squared terms; a hyperbola subtracts them.

Answer Key – CP1

Part 1: Circle & Parabola

#	Task	Answer
1	Rewrite $x^2 + y^2 - 6x + 4y - 612 = 0$ in standard form. Is it the planetarium?	$(x - 3)^2 + (y + 2)^2 = 625$; a circle of radius 25 centered at (3, −2)
2	The satellite dish is 12 ft across and 1.5 ft deep. Find the focal length.	$x^2 = 4py$ through (6, 1.5): $36 = 6p \rightarrow p = 6$ ft
3	How far from the vertex must the receiver be for a dish 10 ft across and 1 ft deep?	$25 = 4p \rightarrow p = 6.25$ ft
4	Write the equation of the flashlight mirror with focus (0, 3) and vertex (0, 0). Find its width at the focus.	$x^2 = 12y$; at $y = 3$, $x = \pm 6 \rightarrow$ 12 units wide

Part 2: Whispering Gallery

#	Task	Answer
5	Write the equation of the ellipse centered at the origin.	$\frac{x^2}{900} + \frac{y^2}{400} = 1$
6	Find the foci. Where should two visitors stand to hear whispers?	$c^2 = 900 - 400 = 500 \rightarrow c \approx 22.4$ ft; at (±22.4, 0)
7	Find the eccentricity. Is the room nearly circular?	$e = \frac{22.4}{30} \approx 0.75$; fairly elongated
8	How far does a whisper travel from one focus to the wall and to the other focus?	$2a = 60$ ft

Part 3: Hyperbolas

#	Task	Answer
9	The cooling tower cross-section is $\frac{x^2}{225} - \frac{y^2}{400} = 1$. Find the narrowest width and the asymptotes.	$2a = 30$ ft; $y = \pm\frac{4}{3}x$
10	Two navigation stations are 100 mi apart at (±50, 0). A ship is 60 mi closer to one than the other. Write the hyperbola.	$2a = 60 \rightarrow a = 30$; $c = 50 \rightarrow b^2 = 1{,}600$; $\frac{x^2}{900} - \frac{y^2}{1{,}600} = 1$
11	Design one more exhibit with a conic. Write its equation and locate its focus or foci.	Answers vary.

Reflection

#	Prompt	Look for
R1	Explain the reflective property of the ellipse in the gallery.	Sound from one focus reflects off the wall to the other focus.
R2	How are a, b, and c related differently in ellipses and hyperbolas?	Ellipse $c^2 = a^2 - b^2$; hyperbola $c^2 = a^2 + b^2$.

Answer Key – Honors

Part 1: Parabolic Engineering

#	Task	Answer
1	Use the definition of a parabola (equal distance from focus (0, p) and directrix y = −p) to derive $x^2 = 4py$.	$\sqrt{(x^2 + (y - p)^2)} = y + p \rightarrow x^2 - 2py = 2py \rightarrow x^2 = 4py$
2	A dish has focal length 4 ft and must be 1.25 ft deep. What diameter should it have?	$x^2 = 16(1.25) = 20 \rightarrow x \approx 4.47 \rightarrow$ diameter ≈ 8.94 ft
3	A parabolic arch over the entrance is 30 ft wide and 18 ft tall. Write its equation with the vertex at the top (0, 18).	$y = 18 - \frac{2}{25}x^2$ (passes through (±15, 0))
4	Can a 12-ft-tall truck that is 10 ft wide pass under the arch? Show your work.	At x = 5, y = 18 − 2 = 16 ft > 12 ft; yes

Part 2: Gallery & Planetarium

#	Task	Answer
5	The museum wants the gallery foci exactly 40 ft apart and the room 50 ft long. Find the width.	a = 25, c = 20 → b = 15 → width 30 ft
6	Find the points where the circle $x^2 + y^2 = 625$ and the ellipse $\frac{x^2}{900} + \frac{y^2}{400} = 1$ intersect.	Substitute $y^2 = 625 - x^2$: $\frac{x^2}{900} + \frac{625 - x^2}{400} = 1 \rightarrow x^2 = 405$ → x ≈ ±20.1, y ≈ ±14.8
7	What does that intersection mean if the planetarium and gallery overlap on a floor plan?	They share four points; the walls would cross there.
8	Find the eccentricity of the new gallery from the first question and compare it to the original.	$e = \frac{20}{25} = 0.8$ vs. ≈ 0.75; the new room is more elongated.

Part 3: Hyperbolic Navigation

#	Task	Answer
9	A ship receives B's signal 0.000322 s before A's. Signals travel 186,000 mi/s. Find the difference in distances and the hyperbola.	$\Delta d \approx 59.9 \approx 60$ mi → $\frac{x^2}{900} - \frac{y^2}{1{,}600} = 1$
10	The ship is also on the line y = 20. Find its position (it is closer to B).	$\frac{x^2}{900} = 1 + \frac{400}{1{,}600} = 1.25 \rightarrow x = 30\sqrt{1.25} \approx 33.5 \rightarrow (33.5, 20)$
11	Write a design memo explaining one conic exhibit, with equations and measurements.	Answers vary.

Reflection

#	Prompt	Look for
R1	How does each conic's geometric definition lead to its useful property?	Parabola: equal distance to focus and directrix → reflection to the focus. Ellipse: constant sum → focus-to-focus reflection. Hyperbola: constant difference → locating by time difference.
R2	Why does eccentricity describe the shape of an ellipse?	It compares the focal distance c with a; as e → 0 the ellipse becomes a circle.